

Reg. No.



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

I SEMESTER M.C.A

MAKE-UP EXAMINATIONS, JANUARY 2024

SUBJECT: OBJECT ORIENTED PROGRAMMING [MCA 4121]

REVISED CREDIT SYSTEM

(08/01/2024)

Time: 3 Hours

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitable assumed.

1A.	A parking garage charges Rs. 20.00 as minimum fee to park for up to three hours. The garage charges an additional Rs. 5.00 for each hour in excess of three hours. The maximum charge for any given 24-hour period is Rs. 100.00 assuming that no car parks for longer than 24 hours at time. Write a C++ program that takes hours parked as input and calculates the parking charges for each of the three customers who parked their cars in this garage yesterday. Create a function called <i>calculate_charge()</i> to perform the above operation. Display the output in the following format: <table> <thead> <tr> <th>Car</th><th>Hours</th><th>Charge</th></tr> </thead> <tbody> <tr> <td>1</td><td>1.5</td><td>20.00</td></tr> <tr> <td>2</td><td>4</td><td>25.00</td></tr> <tr> <td>3</td><td>24</td><td>100.00</td></tr> <tr> <td colspan="2"> </td><td></td></tr> <tr> <td>Total</td><td>29.5</td><td>145.00</td></tr> </tbody> </table>	Car	Hours	Charge	1	1.5	20.00	2	4	25.00	3	24	100.00				Total	29.5	145.00	5
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1B.	Discuss the importance of exception handling in C++ with an example.	3																		
1C.	When do we use the <i>protected</i> visibility specifier in a class?	2																		
2A.	Imagine a publishing company that markets both Book and DVD versions of its works. Create a class <i>Publication</i> that stores the title (a string) and price (type float) of a publication. From this class derive two classes: Book, which adds a page count (type int), and DVD, which adds a playing time in minutes (type float). Each of these three classes should have a <i>getdata()</i> function to get its data from the user, and a <i>putdata()</i> function to display its data. Write a <i>main()</i> program to test the Book and DVD classes by creating object instances respectively, asking the user to fill in data with <i>getdata()</i>, and then displaying the data with <i>putdata()</i>. <i>Hint: Inheritance with function overriding</i>	5																		
2B.	How do <i>pure virtual functions</i> differ from normal <i>virtual functions</i> ? Give a suitable example.	3																		
2C.	Class A : public B, virtual public C{ };	2																		

	In which order are the class constructors called when a derived class object is created?																												
3A.	An employee register contains details of the following workers. Design a program accept employee objects and create a file with these objects. Use object I/O operations on a file. <table><tr><th>Employee id</th><th>Name</th><th>Total sales</th><th>Performance</th></tr><tr><td>1103</td><td>Harvey Specter</td><td>980</td><td>Very Good</td></tr><tr><td>1105</td><td>Micheal Ross</td><td>400</td><td>Poor</td></tr><tr><td>1107</td><td>Glenn Rees</td><td>530</td><td>Average</td></tr><tr><td>1110</td><td>Shane Walsh</td><td>800</td><td>Good</td></tr><tr><td>1112</td><td>Catherine Beckett</td><td>987</td><td>Very Good</td></tr><tr><td>1113</td><td>Catherine Beckett</td><td>867</td><td>Good</td></tr></table> Note: Performance is to be calculated based on the total sales using following conditions: 	Employee id	Name	Total sales	Performance	1103	Harvey Specter	980	Very Good	1105	Micheal Ross	400	Poor	1107	Glenn Rees	530	Average	1110	Shane Walsh	800	Good	1112	Catherine Beckett	987	Very Good	1113	Catherine Beckett	867	Good
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